

LEAKWALL

The Firewall Between Your Secrets and AI

Product Requirements Document & Strategic Roadmap

Universal AI Data Leakage Prevention Platform

Consumers • SMBs • Mid-Market • Enterprise

Version 3.0 — March 17, 2026

CONFIDENTIAL

Budget: \$0 • Team: Solo Founder + Claude • Timeline: 12 Weeks to MVP

1. Executive Summary

LeakWall is an AI data leakage prevention platform that protects individuals and organizations from accidentally exposing sensitive information through AI tools like ChatGPT, Claude, Gemini, and 600+ others. Starting as a free Chrome extension for consumers, it scales through viral bottom-up adoption into paid team and enterprise plans.

The pivot from Kaelus.ai (CMMC compliance for micro-defense contractors) is driven by a brutal market reality: the defense niche has a tiny TAM, impossible trust requirements at \$0 budget, and entrenched competitors. LeakWall targets the single hottest problem in cybersecurity: 77% of employees are leaking company data through ChatGPT, and nobody is selling affordable protection for the 99% of businesses that can't afford \$35K+/year enterprise DLP solutions.

The One-Line Pitch

"Grammarly for data security" — a free Chrome extension that warns you before you paste your company's secrets into ChatGPT.

Key Metrics at a Glance

Metric	Value	Source
Employees leaking data via GenAI	77%	LayerX 2025 Report
Activity on personal/unmanaged accounts	82%	LayerX 2025 Report
GenAI share of unauthorized data movement	32%	LayerX 2025 Report
Shadow AI breach cost increase	+\$670,000	IBM 2025
Average data breach cost	\$4.88 million	IBM 2024
DLP market size (2025)	\$3.4 billion	Fortune Business Insights
DLP market projected (2034)	\$23.76 billion	Fortune Business Insights
Cyberhaven valuation (Apr 2025)	\$1 billion	Series D announcement
GDPR fines in 2025	€1.2 billion	Mordor Intelligence
Employees using GenAI at work	45%	LayerX 2025

2. The Brutal Truth: Why Kaelus Must Pivot

What Killed Kaelus.ai

Problem 1: TAM is too small. 80,000 micro-defense contractors sounds big, but only ~1% are actively pursuing CMMC compliance. That's 800 companies, most of whom won't pay a bootstrapped startup with no SOC 2 certification.

Problem 2: Trust bar is impossible at \$0. Defense contractors need SOC 2 (\$50K-\$200K), FedRAMP authorization, and provable security credentials. You can't fake trust in defense cybersecurity.

Problem 3: Entrenched competitors own the space. PreVeil (\$4,999/year), Drata, Vanta, and Secureframe have raised hundreds of millions. They have sales teams, compliance certifications, and government relationships you can't replicate.

Problem 4: Sales cycle is 6-12 months. Defense procurement moves at government speed. A solo founder with \$0 cannot survive a year of zero revenue waiting for contracts.

What To Keep From Kaelus

Keep (60% reusable)	Kill	Build New
Gateway proxy architecture	CMMC compliance engine	Chrome extension
Scanner library (regex + patterns)	Defense contractor onboarding	Browser paste/upload interception
Dashboard layout + dark theme	ShieldReady assessment wizard	Team management + admin console
Supabase auth + RLS setup	POA&M tracking	Usage analytics + leak reports
Stripe billing integration	NIST 800-171 control database	AI tool detection library
API route structure	Military/defense branding	Consumer-friendly UX

3. Market Proof: Why LeakWall Will Work

The Problem Is Massive and Growing

AI data leakage is not a theoretical risk — it's happening right now, at scale, across every industry. The data is overwhelming:

- 77% of employees paste company data into GenAI tools, with 82% doing so through personal accounts that bypass all corporate controls (LayerX 2025)
- GenAI is now the #1 data exfiltration channel in enterprises, responsible for 32% of all unauthorized data movement — surpassing email and cloud storage for the first time (Cyera 2025)
- Among users who paste into GenAI, the average is 6.8 pastes per day, with 3.8 containing sensitive corporate data (LayerX 2025)
- 40% of files uploaded to GenAI tools contain personally identifiable information or payment card data (LayerX 2025)
- 59% of employees use unapproved AI tools at work, and 75% of those share sensitive information (Cybernews 2025)
- Shadow AI usage increases the average cost of a data breach by \$670,000 (IBM 2025)
- In January 2026, two Chrome extensions with 900,000+ downloads were caught stealing conversations from ChatGPT and DeepSeek (The Hacker News)

The Market Is Exploding

The DLP market is growing at 22-24% annually, driven by GenAI adoption and tightening regulations:

- DLP market: \$3.4B (2025) projected to \$23.76B by 2034 at 24.1% CAGR (Fortune Business Insights)
- Broader DLP including managed services: \$33.26B (2025) to \$111.98B by 2031 at 21.17% CAGR (Mordor Intelligence)
- Cyberhaven reached \$1B valuation with \$100M Series D in April 2025 — a 7x increase in just one year
- Palo Alto Networks acquired CyberArk for \$25B in February 2026, fusing access control with DLP
- Zscaler acquired SPLX to embed natural-language data classification across Slack and Notion
- GDPR fines hit €1.2 billion in 2025, up 22% year-over-year

The Gap Nobody Is Filling

There is a massive pricing vacuum between enterprise DLP tools and open-source solutions:

Segment	Solution	Price	Problem
Enterprise	Cyberhaven, Palo Alto, Zscaler	\$35K-100K+/year	Requires security team, 6-month deployment

Segment	Solution	Price	Problem
Mid-market	Nightfall AI, Code42	\$14-52/user/month	Still expensive, complex setup
SMB	???	???	NOTHING EXISTS
Consumer	Lakera (basic)	Free	ChatGPT only, no team features, limited
LeakWall	Chrome ext → Dashboard	\$0-19/user/month	The entire missing middle

4. Product Specification

4.1 Chrome Extension (V1 — Free Tier)

The Chrome extension is the core product and primary distribution vehicle. It runs entirely in the browser with zero server dependency for basic functionality.

Detection Engine

Content script intercepts paste events, keystrokes, and file uploads on AI tool pages. Pattern matching runs locally in the browser:

Data Type	Detection Method	Example Pattern
Social Security Numbers	Regex + Luhn validation	XXX-XX-XXXX format
Credit Card Numbers	Regex + Luhn algorithm	Visa, MC, Amex patterns
API Keys / Tokens	Entropy analysis + prefix matching	sk_live_, ghp_, AKIA...
Email Addresses	RFC 5322 regex	user@domain.tld
Phone Numbers	International format regex	+1-XXX-XXX-XXXX
Source Code	Keyword + structure detection	function, import, class, def
Medical Record Numbers	Format matching	MRN patterns
Financial Data	Currency + number patterns	\$XX,XXX.XX
IP Addresses	IPv4/IPv6 regex	192.168.x.x
Passwords	Context detection	"password": "..."

Supported AI Tools (V1: Top 10)

Tool	URL Pattern	Monthly Users
ChatGPT	chat.openai.com, chatgpt.com	300M+
Claude	claude.ai	30M+
Gemini	gemini.google.com	50M+
Microsoft Copilot	copilot.microsoft.com	40M+
DeepSeek	chat.deepseek.com	20M+
Perplexity	perplexity.ai	15M+
Poe	poe.com	10M+
Jasper	jasper.ai	5M+
Copy.ai	app.copy.ai	3M+
Mistral	chat.mistral.ai	5M+

4.2 Team Dashboard (V2 — Paid Tier)

The admin dashboard provides team-level visibility into AI data leakage patterns without exposing individual content:

- Leak event feed: severity, data type, AI tool, timestamp (content never stored)
- Policy controls: block vs warn, custom rules, sensitivity thresholds
- Team analytics: leaks by department, tool, data type, trend over time
- Compliance exports: PDF/CSV reports for SOC 2, GDPR, HIPAA evidence
- Member management: invite, remove, assign roles (admin/viewer)

4.3 Safe AI Mode (V3 — Premium Feature)

Built-in AI redaction proxy that strips sensitive data before it reaches any LLM:

- User types prompt containing sensitive data → LeakWall intercepts
- Sensitive data replaced with safe placeholders: [SSN_1], [API_KEY_1], etc.
- Sanitized prompt sent to the AI tool
- AI response received → placeholders mapped back to original values in user's browser
- Net result: full AI productivity, zero data exposure to the LLM provider

5. Pricing Strategy

Tier	Price	Target Segment	Key Features
Free	\$0	Individuals	Extension, basic warnings, 5 AI tools, local-only
Pro	\$5/month	Freelancers, power users	Unlimited AI tools, custom rules, export history
Team	\$9/user/month	SMBs (5-50 people)	Admin dashboard, team analytics, policy controls
Business	\$19/user/month	Mid-market (50-500)	SSO, advanced policies, compliance reports, API
Enterprise	Custom	500+ employees	Dedicated support, custom integrations, SLA, on-prem

Pricing Rationale

- Free tier is the distribution engine — it must be genuinely useful, not a teaser
- \$5 Pro creates first revenue signal and validates willingness to pay
- \$9/user Team is 50-90% cheaper than any competitor with team features
- \$19/user Business still undercuts Nightfall (\$25/user) and Code42 (\$30/user)
- Enterprise pricing determined by sales conversations once product-market fit is proven

6. Technical Architecture (\$0 Budget Stack)

Component	Technology	Cost	Free Tier Limits
Frontend	Next.js 15 + React 19 + Tailwind	\$0	N/A (open source)
Chrome Extension	Manifest V3, TypeScript	\$5 one-time	Chrome Web Store fee
Backend	Next.js API Routes + Supabase Edge Functions	\$0	500K invocations/month
Database	Supabase PostgreSQL	\$0	500MB, 50K MAU
Auth	Supabase Auth	\$0	50,000 MAU
Payments	Stripe	\$0 until revenue	2.9% + \$0.30 per transaction
Hosting	Vercel	\$0	100GB bandwidth, 100hrs compute
AI (premium features)	Claude API via OpenRouter	\$0 until needed	Pay per token
Analytics	PostHog	\$0	1M events/month
Email	Resend	\$0	100 emails/day

Total launch cost: \$5 (Chrome Web Store developer registration fee)

7. Competitive Landscape

Company	Valuation/Revenue	Target	Pricing	LeakWall Advantage
Cyberhaven	\$1B valuation	Fortune 500	\$35K+/year	Free tier, no deployment needed
Palo Alto Prisma	\$120B market cap	Enterprise	\$50K+/year	Browser-native, no stack needed
Zscaler	\$30B market cap	Enterprise	\$30K+/year	Sees paste events, not just network
Nightfall AI	\$100M+ raised	Mid-market	\$25/user/month	50% cheaper, consumer entry point
Code42 Incydr	Acquired \$500M	Mid-market	\$30/user/month	AI-specific, not general DLP
Lakera	\$20M raised	Developers	Free extension	Multi-tool, team features, dashboard

The key competitive insight: every player above sells top-down to enterprises. None of them can or will build a free Chrome extension that goes viral with individuals and pulls teams/companies upward through bottom-up adoption. This is not a technology moat — it's a distribution and business model moat.

8. Distribution Strategy: The Grammarly Playbook

LeakWall follows the proven bottom-up SaaS distribution model used by Grammarly (700M+ users), Honey (acquired by PayPal for \$4B), LastPass, 1Password, and Slack:

The Viral Loop

1. Individual installs free Chrome extension for personal AI privacy
2. Extension catches sensitive work data → user realizes it's a work problem too
3. User tells manager: "We need this for the whole team"
4. Manager buys Team plan (\$9/user/month) for department
5. IT/Security evaluates for company-wide deployment → Business/Enterprise tier

Launch Distribution Channels (\$0 Budget)

Channel	Action	Expected Impact	Timeline
Reddit	r/privacy, r/cybersecurity, r/ChatGPT, r/sysadmin	500-2,000 installs	Week 7
Hacker News	Show HN: LeakWall — free AI data leak protection	1,000-5,000 installs	Week 7
Product Hunt	Full launch with demo video	500-3,000 installs	Week 8

Channel	Action	Expected Impact	Timeline
LinkedIn	"77% of your employees are leaking data" posts	200-1,000 installs	Weeks 7-12
Twitter/X	AI security thread with free tool CTA	100-500 installs	Weeks 7-12
YouTube	"I built a tool that stops ChatGPT data leaks" video	200-500 installs	Week 9
Indie Hackers	Build-in-public journey + launch post	100-300 installs	Week 8

9. Detailed 12-Week Build Roadmap

Phase 0: Foundation (Week 1)

- Fork Kaelus.ai codebase to new LeakWall repository
- Strip all CMMC-specific code (ShieldReady, NIST controls, POA&M, defense branding)
- Rebrand: new logo, color palette, landing page copy
- Set up Chrome extension scaffold (Manifest V3, content script, popup, service worker)
- Configure Supabase tables: leak_events, users, teams, policies
- Update environment variables for new brand

Deliverable: Clean codebase, extension skeleton loads in Chrome, database ready

Phase 1: Core Extension Engine (Weeks 2-4)

- Week 2: Content script injected on AI tool pages, intercepts paste/type events
- Week 2: Pattern matching engine V1 — SSN, credit cards, API keys, emails
- Week 3: Source code detection (function signatures, imports, class definitions)
- Week 3: Extension popup UI — toggle protection, sensitivity slider, recent alerts
- Week 4: Support 10 AI tools (ChatGPT, Claude, Gemini, Copilot, DeepSeek, etc.)
- Week 4: Warning overlay UX — non-intrusive toast with "Block" / "Allow" / "Redact" options

Deliverable: Working extension that catches and warns about sensitive data across 10 AI tools

Phase 2: Ship Free Version (Weeks 5-6)

- Week 5: Polish extension UX, smooth onboarding (3-step welcome flow)
- Week 5: Landing page (reuse Next.js from Kaelus, new copy and design)
- Week 5: Privacy policy, terms of service, security whitepaper
- Week 6: Chrome Web Store submission (allow 3-5 days for review)
- Week 6: PostHog analytics (anonymized — event types only, never user content)
- Week 6: Automated tests, edge case handling, performance optimization

Deliverable: Live Chrome extension available for public download

Phase 3: Distribution Blitz (Weeks 7-8)

- Week 7: Reddit posts — r/privacy, r/cybersecurity, r/ChatGPT, r/sysadmin, r/artificial
- Week 7: Hacker News Show HN submission
- Week 7: LinkedIn content campaign (3 posts/week, data-driven angles)
- Week 8: Product Hunt launch
- Week 8: Indie Hackers build-in-public post
- Week 8: Collect user feedback, prioritize feature requests

Target: 1,000 free installs by end of Week 8

Phase 4: Monetization (Weeks 9-10)

- Week 9: Admin dashboard — team leak event feed, policy controls, member management
- Week 9: Team onboarding flow — invite members via email, bulk extension deploy
- Week 10: Stripe integration — Pro (\$5/mo) and Team (\$9/user/mo) checkout
- Week 10: Email alerts for admins on high-severity leak events
- Week 10: Usage-based upgrade prompts in free extension

Target: First 10 paying accounts, \$500-2,000 MRR

Phase 5: Iterate and Scale (Weeks 11-12)

- Week 11: Respond to user feedback, fix bugs, add top-requested AI tools
- Week 11: Blog content — "How [industry] companies protect against AI data leaks"
- Week 12: Direct outreach to 50 SMBs from Reddit/LinkedIn who showed interest
- Week 12: Collect case studies from first paying customers
- Week 12: Plan Phase 6 features based on real user data

Target: 2,500+ free users, 20+ paying accounts, clear signal on product-market fit

Phase 6: Growth (Months 4-12)

- Safe AI Mode: redaction proxy for premium users (strip data before sending to LLM)
- Compliance reports: SOC 2, GDPR, HIPAA evidence generation
- Firefox extension (expand browser coverage)
- API for custom integrations
- Business tier (\$19/user/month) with SSO, advanced policies
- Content marketing engine: weekly blog, monthly security reports

Target: 100 team accounts, \$10K+ MRR, clear path to \$100K ARR

10. Revenue Projections

Timeline	Free Users	Paid Users	Avg. Revenue/User	MRR	ARR
Month 1-2	500-1,000	0	\$0	\$0	\$0
Month 3	2,000	20	\$9	\$180	\$2,160
Month 6	5,000	100	\$9	\$900	\$10,800
Month 9	10,000	300	\$9	\$2,700	\$32,400
Month 12	25,000	800	\$9	\$7,200	\$86,400
Month 18	50,000	2,000	\$9	\$18,000	\$216,000
Month 24	100,000	5,000	\$9	\$45,000	\$540,000

These projections assume a conservative 2-3% free-to-paid conversion rate, consistent with industry benchmarks for freemium security tools. Grammarly converts at ~5%, LastPass at ~4%. Given the urgency of AI data leakage and growing regulatory pressure, 3% conversion is achievable.

Revenue Milestones

- First dollar: Month 3 (10 weeks after starting development)
- Ramen profitable (\$2,000 MRR): Month 6-9
- Real business (\$10K MRR / \$120K ARR): Month 12-18
- Venture-scale signal (\$50K MRR / \$600K ARR): Month 24
- Acquisition-attractive (\$100K+ MRR / \$1.2M+ ARR): Month 30-36

11. Risk Assessment

Risk	Probability	Impact	Mitigation Strategy
Big player builds same thing	Medium	High	Speed to market, community moat, bottom-up distribution they can't replicate
Chrome Web Store rejection	Low	High	Follow Manifest V3 strictly, minimal permissions, clear privacy policy
Low free-to-paid conversion	Medium	Medium	Team features create natural upgrade; A/B test pricing and prompts
Solo founder burnout	High	High	Ship MVP in 4 weeks, validate before building more; say no to scope creep
Privacy concerns about extension	Medium	High	Open-source extension code, zero-knowledge architecture, all processing local
Pattern matching false positives	High	Medium	User feedback loop, adjustable sensitivity, per-site whitelists

Risk	Probability	Impact	Mitigation Strategy
Regulatory changes to extensions	Low	Medium	Manifest V3 compliant from day 1; diversify to Firefox early
AI tools block extension	Low	High	Extension runs at browser level, not API level; hard to detect

12. Moat and Defensibility

Five-Layer Moat

- 1. Network Effect:** More users generate more pattern data, improving detection accuracy for everyone. Each new AI tool supported makes the extension more valuable for all users.
- 2. Data Advantage:** Anonymized leak pattern data (never content) creates a proprietary understanding of what data types flow into which AI tools, informing better detection models.
- 3. Switching Costs:** Teams configure custom policies, train employees on the workflow, build compliance processes around LeakWall reports. Switching means re-training everyone.
- 4. Brand Trust:** In security, trust compounds over time. Being the first privacy-first, open-source option builds brand equity that late entrants can't buy.
- 5. Bottom-Up Distribution:** Enterprise competitors sell top-down through 6-month sales cycles. They structurally cannot replicate viral individual-to-team-to-company adoption.

13. Exit Scenarios

Scenario	Timeline	Valuation Range	Likely Acquirers
Acqui-hire	Year 1-2	\$1-5M	Security startups needing browser-native DLP
Strategic acquisition	Year 2-3	\$20-100M	Cyberhaven, Nightfall, Code42, CrowdStrike
Major acquisition	Year 3-5	\$100-500M	Palo Alto, Zscaler, Microsoft, Google
AI company acquisition	Year 2-4	\$50-200M	OpenAI, Anthropic (enterprise security layer)
IPO path	Year 5+	\$500M+	If \$50M+ ARR achieved
Lifestyle business	Ongoing	\$1-5M profit/year	Solo founder, 50-70% margins

The AI data security space is in a period of aggressive M&A consolidation. Six major acquisitions occurred in 2024-2026 alone (Cyberhaven Series D, Zscaler/SPLX, Zscaler/Red Canary, Palo Alto/CyberArk, Fortra/Lookout, CrowdStrike Falcon Data Protection). Building a product with meaningful traction in this space makes acquisition highly likely within 2-3 years.

14. Your Exact Next Steps

This is the most important section. Everything above is strategy. This is action.

Today (March 17, 2026)

6. Save the mega-prompt somewhere permanent (Google Docs, Notion, local file)
7. Fork the Kaelus.ai repo to a new "leakwall" repository
8. Start Phase 0: strip CMMC code, rebrand, scaffold Chrome extension

This Week (Week 1)

9. Complete Phase 0 — clean codebase, extension loads in Chrome
10. Register Chrome Web Store developer account (\$5)
11. Buy leakwall.dev or leakwall.io domain (optional, use Vercel subdomain if \$0)

By End of Month (Weeks 2-4)

12. Working extension that catches sensitive data on ChatGPT, Claude, and Gemini
13. Pattern matching for SSNs, credit cards, API keys, and source code
14. Extension popup with toggle, sensitivity slider, and alert history

The window is open right now. AI data leakage is headline news, regulators are cracking down, enterprises are panicking, and nobody is serving the 99% who can't afford \$35K/year DLP tools. Ship fast.

Appendix A: Data Sources and Citations

- LayerX Security, Enterprise AI and SaaS Data Security Report 2025
- IBM, Cost of a Data Breach Report 2024/2025
- Cybernews, "59% of employees use unapproved AI tools at work" (Sept 2025)
- eSecurity Planet, "77% of Employees Leak Data via ChatGPT" (Oct 2025)
- The Hacker News, "AI is Already the #1 Data Exfiltration Channel" (Oct 2025)
- The Hacker News, "Two Chrome Extensions Caught Stealing AI Chats" (Jan 2026)
- Cyberhaven, Series D announcement, \$1B valuation (Apr 2025)
- Cyberhaven, FY2026 Growth Report (Feb 2026)
- Cyberhaven Labs, 2026 AI Adoption & Risk Report (Feb 2026)
- Fortune Business Insights, Data Loss Prevention Market Report (2025-2034)
- Mordor Intelligence, Data Loss Prevention Market Report (2025-2031)
- Precedence Research, Data Loss Prevention Market (2025-2034)
- Palo Alto Networks / CyberArk acquisition (\$25B, Feb 2026)
- Zscaler / SPLX acquisition (Nov 2025)
- Tom's Guide, "Employees are unknowingly leaking company secrets through ChatGPT" (Oct 2025)

Appendix B: Glossary

- DLP: Data Loss Prevention — technologies that detect and prevent unauthorized transmission of sensitive data
- GenAI: Generative AI — AI systems that generate text, images, or code (ChatGPT, Claude, Gemini, etc.)
- Shadow AI: Use of unapproved AI tools within an organization without IT oversight
- PII: Personally Identifiable Information — data that can identify a specific individual
- PCI: Payment Card Industry — credit card numbers and related financial data
- GDPR: General Data Protection Regulation — EU data privacy law
- SOC 2: Service Organization Control 2 — trust service criteria for data security
- MRR: Monthly Recurring Revenue
- ARR: Annual Recurring Revenue
- MAU: Monthly Active Users
- CTA: Call To Action